

The Exchange Phase as a Logical Scalar: $R = e^{2\pi i s}$ from the Re-Entrant Calculus

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Abstract

The boson/fermion dichotomy is conventionally presented as a primitive classification of nature. Recent work established that the underlying invariant is the relation between the exchange phase of identical particles and their topological spin, $R = e^{2\pi i s}$, with the dichotomy as its shadow in three spatial dimensions where the involutive braiding quantizes the spin parameter. This paper examines whether a calculus whose only primitive is the act of drawing a distinction can generate this invariant rather than importing it as an axiom. The re-entrant mark of the calculus of indications, disciplined by linearity, already generates the constants e and π as logical scalars: e as the fixed point of the differential equation $Df = f$, and π as the trace of the identity on the circle type. The central observation of this paper is that the exchange phase is the $(2s)$ -fold half-turn of the re-entrant mark: $R = (e^{i\pi})^{2s} = e^{2\pi i s} = (-1)^{2s}$, so that the boson/fermion dichotomy is the parity of $2s$. The arithmetic identity is established; the identification of the exchange monodromy with a power of the mark's half-turn is a model of the re-entrant phase; the claim that the calculus derives the invariant as a logical scalar within a single formal system is stated as a conjecture with explicit falsifiability conditions. The framework is distinguished from the nearest prior work deriving a Z_2 exchange phase from self-referential scattering, which lacks both the power structure for arbitrary spin and the unification of e , π , and R as a single scalar family.

Keywords: exchange phase; spin-statistics; topological spin; laws of form; re-entrant mark; logical scalar; anyons

1. Introduction

The spin-statistics theorem is among the most robust laws of physics: integer-spin particles obey Bose–Einstein statistics and half-integer-spin particles obey Fermi–Dirac statistics [Pauli 1940]. The QNFO research program has pursued the structural reading of this law. In [Quni-Gudzinis 2026b] it was established that the primitive content is not the dichotomy itself but the relation

$$R = e^{2\pi i s} \quad [\text{ESTABLISHED}]$$

between the exchange phase of identical particles and their topological spin, where the boson/fermion binary is a dimension-dependent shadow of this relation in three spatial dimensions [ESTABLISHED]. The same work identified a derivation target: a distinction-based foundation of physics must derive exchange statistics from the primitive act of drawing a boundary, rather than importing the relation as an axiom. The required construction was formalized — two modal exponentials, the braiding of two marks in a compact closed category, and the ribbon condition — yielding $\eta = \pm 1$ in the symmetric case, with $\eta = -1$ identified with the treatise’s half-turn phase $e^{\{i\pi\}} = -1$.

This paper closes the remaining gap with one structural observation: **the exchange phase is the (2s)-fold half-turn of the re-entrant mark**. The constants e and π have already been shown to arise as logical scalars of the re-entrant calculus — values the type theory computes rather than axioms it imports [Quni-Gudzinis 2026a]. This paper extends the scalar family to the exchange phase $R = (e^{\{i\pi\}})^{\{2s\}}$, and states precisely what is established, what is a model, and what is conjectural about that identification.

2. Background: the invariant and the machinery

2.1 The invariant $R = e^{\{2\pi i s\}}$

The exchange of two identical particles is a loop in their configuration space. The phase acquired is a representation of the fundamental group of that space: $\pi_1 = \mathbb{Z}_2$ in dimension $d \geq 3$, and $\pi_1 = B_2 \cong \mathbb{Z}$ in dimension $d = 2$ [Leinaas and Myrheim 1977]. The spin-statistics relation states which phase is realized: $\eta = (-1)^{\{2s\}}$, where s is the spin [Pauli 1940; Duck and Sudarshan 1998]. In dimension $d \geq 3$ the involutive braiding forces $2s \in \mathbb{Z}$, so the exchange phase is ± 1 and particles are bosons or fermions; in dimension $d = 2$ the quantization collapses to continuity, and arbitrary exchange phases $e^{\{2\pi i s\}}$ describe anyons [Wilczek 1982; Kitaev 2006]. [ESTABLISHED]

2.2 The re-entrant calculus

The calculus of indications [Spencer-Brown 1969] takes the drawing of a distinction as primitive. The re-entrant form $f = \bar{f}$ oscillates, generating a discrete clock of period 2: $f(n) = (-1)^n f(0)$ [ESTABLISHED — elementary consequence]. Under linear discipline [Quni-Gudzinis 2026a]:

- e arises as the logical scalar of the differential fixed point: $D f = f$, $f(0) = 1$, with unique solution $f(x) = e^x$ and $e = \sum 1/n!$ [established analysis];
- π arises as the trace of the identity on the circle type: $\pi = \text{Tr}(\text{id}_{\{S^1\}}) = C/d$ in the analytic realization, where the circle carries its geometric structure and the trace is computed by the Gaussian-integral construction [established geometry; logical derivation: [my conjecture]];
- **the half-turn** of the circle carries the marked state to the unmarked state: $e^{\{i\pi\}} = -1$, the geometric root of the Euler identity [ESTABLISHED — Euler’s formula].

The exchange of two marks in a compact closed category was constructed in [Quni-Gudzinis 2026b, notebook T2]: the exchange map $\sigma_{\{M,M\}} = \eta \cdot \text{id}$ for the self-dual mark, the ribbon identity $\eta = \theta_M$, and the symmetric-category constraint $\eta = \pm 1$, with $\eta = -1$ identified with Crossing ($e^{\{i\pi\}} = -1$) and $\eta = +1$ with Calling.

2.3 The gap

The parent works establish $\eta = \pm 1$ (symmetric case) and identify the single sign $\eta = -1$ with the treatise's half-turn, but never write the power structure $R = (e^{\{i\pi\}})^{\{2s\}}$ that covers arbitrary spin s and the 2+1-dimensional anyon generalization. The leap from “the mark has a half-turn phase” to “two marks anticommute” is asserted, not derived [Quni-Gudzinis 2026b]. This paper supplies the missing composite reading.

3. Core claim

The exchange phase $R = e^{\{2\pi i s\}}$ is the $(2s)$ -fold half-turn of the re-entrant mark:

$$R = (e^{i\pi})^{2s} = e^{2\pi i s} = (-1)^{2s},$$

with the boson/fermion dichotomy the parity of $2s$.

Formally: - $s \in \mathbb{Z}$: $2s$ even $\rightarrow R = +1 \rightarrow$ Bose–Einstein statistics (symmetric exchange, Calling); - $s \in \mathbb{Z} + \frac{1}{2}$: $2s$ odd $\rightarrow R = -1 \rightarrow$ Fermi–Dirac statistics (antisymmetric exchange, Crossing); - 3+1-dimensional involutive braiding quantizes s to $\{0, 1/2, 1, 3/2, \dots\} \rightarrow$ the dichotomy [ESTABLISHED, Quni-Gudzinis 2026b]; - 2+1 dimensions allow any real $s \rightarrow$ anyon phases $e^{\{2\pi i s\}}$ [ESTABLISHED, Leinaas and Myrheim 1977].

Scope note on the equality chain. The final equality $(-1)^{\{2s\}} = \pm 1$ holds only in the quantized case $2s \in \mathbb{Z}$. For arbitrary real s the general form is $R = e^{\{2\pi i s\}} = \cos 2\pi s + i \sin 2\pi s$ (e.g., $s = 1/4$ gives $R = i$). The specific reading advanced here is the middle form: R as a power of the treatise's half-turn, $(e^{\{i\pi\}})^{\{2s\}}$ — the monodromy of the exchange loop as the $(2s)$ -fold iteration of the half-turn $e^{\{i\pi\}} = -1$.

4. The derivation

Step 1 — the exchange map exists in the compact closed structure. [established — construction] The exchange $\sigma_{\{M,M\}}: M \otimes M \rightarrow M \otimes M$ is the braiding of the self-dual mark. In a symmetric braided category $\sigma^2 = \text{id}$, so σ has eigenvalues ± 1 with idempotent projectors $P_{\text{sym}} = \frac{1}{2}(1+\sigma)$ and $P_{\text{antisym}} = \frac{1}{2}(1-\sigma)$.

Step 2 — the half-turn as the basic monodromy. [established — treatise §12.1] The half-turn of the circle carries the marked state to the unmarked state: $e^{\{i\pi\}} = -1 = \text{Crossing}$. This is the single-mark monodromy under rotation by π .

Step 3 — exchange phase as a power of the half-turn. [MAP — model of the re-entrant phase] The exchange of two particles is a loop in their configuration space. The loop winds the relative coordinate; the phase acquired is the monodromy of that loop. The reading advanced here is that the exchange monodromy is the $(2s)$ -fold

iteration of the mark’s half-turn: $R = (e^{\{i\pi\}})\{2s\} = e^{\{2\pi i s\}} = (-1)^{\{2s\}}$. The arithmetic identity is [established]; the identification of the exchange monodromy with a power of the mark’s half-turn is [MAP — model].

Step 4 — parity of 2s → the dichotomy. [established arithmetic + Quni-Gudzinās 2026b §1] The dichotomy is exactly the parity of 2s.

Step 5 — dimension quantization. [established — Quni-Gudzinās 2026b §3] In $d \geq 3$ the involutive braiding forces $2s \in \mathbb{Z} \rightarrow R = \pm 1$; in $d = 2$ the braid group allows continuous $s \rightarrow$ anyon phases $e^{\{2\pi i s\}}$.

Step 6 — unification with e and π. [my conjecture — the scalar family] e (the fixed point of $D f = f$), π (the trace of the identity on S^1), and R (the monodromy power of the half-turn) form one family of logical scalars of the re-entrant mark under linear discipline: fixed point, trace, monodromy power.

5. Status ladder

Component	Status
Exchange map $\sigma_{\{M,M\}}$; projectors $P_{\text{sym}}/P_{\text{antisym}}$	[established — Quni-Gudzinās 2026b, notebooks T1/T2]
Half-turn $e^{\{i\pi\}} = -1$	[established — treatise §12.1]
$\eta = -1 \leftrightarrow$ Crossing; $\eta = +1 \leftrightarrow$ Calling	[established — Quni-Gudzinās 2026b, notebook T2]
$(e^{\{i\pi\}})\{2s\} = (-1)^{\{2s\}} = \pm 1$ for $2s \in \mathbb{Z}$	[established — elementary arithmetic]
Exchange monodromy = (2s)-fold half-turn	[MAP — model of the re-entrant phase]
$e/\pi/R$ scalar-family unification	[my conjecture]
Formal derivation in the traced differential cohesive linear type theory	[my conjecture — F1 target]
Physical realization (which sign, in 3+1D)	[established physics; external Lorentz/microcausality input]

6. Falsifiability conditions

- F1 (formal).** The claim that the re-entrant calculus *generates* $R = (e^{\{i\pi\}})\{2s\}$ as a logical scalar is falsified if no derivation exists within the traced differential cohesive linear type theory of the treatise (Part VIII) without importing the relation as an axiom. This is a concrete, checkable claim about a formal system. [my conjecture]
- F2 (empirical, inherited).** If a stable, local, relativistic excitation in 3+1 dimensions is observed with exchange phase $\eta \neq e^{\{2\pi i s\}}$ (e.g., a spin-1/2 particle obeying Bose–Einstein statistics), the invariant claim is disconfirmed. No such particle is known in the Standard Model. Evasion strategies in the literature (e.g., mass-dimension-three-half spinors [Ahluwalia and Lee 2022]) target the standard theorem statement rather than the invariant relation itself. [established — restated from Quni-Gudzinās 2026b F1]

- **F3 (scope).** The arithmetic identity $R = (e^{\{i\pi\}})\{2s\} = (-1)^{\{2s\}}$ is [established] elementary arithmetic; the identification of the exchange phase with the (2s)-fold half-turn in the geometric model is [MAP — model]; the claim that this identification is a logical derivation within a single formal system is [my conjecture].

Disconfirmation summary. The invariant claim is disconfirmed if: (a) a formal derivation in the Part VIII system is shown impossible without importing the relation as an axiom (F1); or (b) an empirical excitation with exchange phase $\eta \neq e^{\{2\pi i s\}}$ is observed in 3+1 dimensions (F2). Neither condition is currently met.

7. Relation to prior work

Quni-Gudzinis 2026b (parent). Establishes the invariant $R = e^{\{2\pi i s\}}$ and the $\eta = \pm 1$ symmetric construction; explicitly documents that the leap from the half-turn phase to anticommuting marks is not derived. This paper supplies the $(e^{\{i\pi\}})\{2s\}$ composite reading absent there.

Kauffman 2022. Reviews Majorana fermions through the laws of form, connecting the mark calculus to fermionic structure. It addresses the representation of fermions, not the exchange-phase invariant, and contains no (2s)-fold half-turn power structure. The present claim is distinct and complementary.

Ma and Zhang 2025. Derive a Z_2 exchange phase from self-referential scattering via Riccati square roots and the spinor double cover in a quantum field theory framework. Their primitive is self-referential scattering in QFT; the primitive here is the re-entrant mark of the calculus of indications. Their result is confined to the Z_2 (boson/fermion) case; the $(e^{\{i\pi\}})\{2s\}$ power structure for arbitrary s (anyons) and the $e/\pi/R$ scalar-family unification are absent. The present claim is distinct: the derivation target is the re-entrant mark under linear discipline, not bare self-reference.

Berry and Robbins 2017. The geometric-phase construction of spin-statistics: exchange of two particles acquires the Berry phase, connecting statistics to geometry. The monodromy-power reading advanced here is conceptually adjacent (both trace the exchange phase to a geometric monodromy) but is formulated natively in the calculus of indications rather than in Hilbert-space geometry.

Ahluwalia and Lee 2022. Propose mass-dimension-three-half spinors as an evasion of the standard spin-statistics theorem. This is relevant to F2: the evasion targets the standard theorem statement, not the invariant R itself; the empirical falsifier above covers the general evasion class.

8. Conclusions

The exchange phase $R = e^{\{2\pi i s\}}$ is the (2s)-fold half-turn of the re-entrant mark: $R = (e^{\{i\pi\}})\{2s\} = (-1)^{\{2s\}}$. The boson/fermion dichotomy is the parity of 2s. The claim is scoped with an explicit status ladder: the arithmetic is established, the identification is a model, and the full logical derivation within the traced differential cohesive linear type theory of the treatise is a conjecture with concrete falsifiability.

conditions. If the derivation succeeds, e , π , and R form a single family of logical scalars of the re-entrant mark — fixed point, trace, and monodromy power — and the spin-statistics connection is the arithmetic of the half-turn. If it fails, the failure mode is precisely specified: the calculus cannot generate the exchange phase without importing it as an axiom.

9. Declarations

- **Funding:** This research received no external funding.
- **Conflicts of interest:** The author declares no conflicts of interest.
- **Data availability:** All source files, gate artifacts, and external-search evidence are deposited with this record and mirrored in the project repository (see provenance link).
- **Code availability:** The derivation is analytical; the arithmetic verification and citation-audit scripts are deposited as artifacts.
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- **Preprint policy:** This is a self-archived working paper; it has not been submitted for peer review.
- **Reproducibility:** Every numerical claim is independently recomputable from the stated elementary formulas; the citation audit re-verifies every bibliographic entry against live registries.

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